

Africa Specialty Risks

Enabling Development

Flood Risk Analysis

June 2026

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Definition:



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A flood is a temporary overflow of water onto land that is normally dry. It occurs when the **Inflow** (precipitation/river discharge) exceeds the **Loss** (soil infiltration/drainage capacity).

Flood = (P - I) > C

Where P is Precipitation (Intensity/Duration), I is Infiltration (Soil Saturation/Percolation), and C is Channel/Drainage Capacity.

In **Arid/Semi-Arid regions** (e.g., Sahel): A flood occurs when rainfall intensity exceeds the **infiltration rate** of parched, compacted soils, leading to high-velocity runoff (Flash Flooding).

In **Humid/Tropical regions** (e.g., Central Africa): A flood occurs when cumulative rainfall exceeds the **soil storage capacity**, leading to a 'saturation overland flow' where the ground can no longer absorb water regardless of intensity."

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Pluvial Flood / Urban Flood

- An Urban Flood is defined by **Rainfall Intensity** (how much falls in a very short time) because the concrete (Impervious Surface) cannot drain it fast enough.
- An event where 24-hour rainfall exceeds the WMO R95p **95th Percentile** of the historical daily rainfall (calculated over a 30-year baseline) for that specific district.
- *Following **Panthou et al. (2018)** and **Douglas et al. (2008)** regarding African urban centres, we define an Urban Flood as a rainfall event exceeding the 95th percentile of local historical intensity, which triggers 'pluvial' inundation due to the 10% impervious cover tipping point (Schueler, 2009).*
- **World Bank (2021) "Urban Flood Risk Management in the Pacific/Africa."** It states that urban flood definitions must be based on *intensity thresholds* rather than cumulative weekly totals.

Calculation

The Formula: The R95p Threshold

$$Flood_{Urban} = \begin{cases} 1, & \text{if } P_{24h} > P_{95th} \text{ AND } ISA > 10\% \\ 0, & \text{otherwise} \end{cases}$$

- **P_{24h}:** The total rainfall measured in a 24-hour window.
- P_{95th}**:** The 95th percentile of historical daily rainfall (the "Heavy Precipitation" threshold).
- ISA**:** Impervious Surface Area (ESA Class 50).
- **WMO Expert Team on Climate Change Detection and Indices (ETCCDI)** standard.
- It identifies "Flash" events. In a city, if the rain falls in the top 5% of historical intensity, the pipes and gutters (designed for average rain) will overflow.

Agricultural/Rural Flood

- A Rural Flood is defined by **Accumulation and Saturation** (how much falls over many days). The ground acts like a sponge; it only floods once the sponge is full.
- An event where the **7-day cumulative rainfall** exceeds **200% historical standard deviation** for that week, OR where rainfall exceeds the **Soil Infiltration Capacity**.
- This is called **"Saturation Excess."** Crops don't die from one heavy rain; they die because the water sits on the field for 3–7 days because the ground is already saturated.
- *Following **Tramblay et al. (2021)** on 'Flood-generating processes in Africa,' we define Agricultural Flooding as a 'long-rain' event where cumulative 7-day precipitation exceeds 2.0x the historical standard deviation, leading to saturation-excess runoff.*

Calculation

The Formula: Runoff Depth (Q)

$$Q = \frac{(P - I_a)^2}{(P - I_a + S)}$$

The Agriculture Trigger:

$$Flood_{Ag} = 1 \text{ if } \sum P_{7d} > (2 \times \sigma) \text{ AND } Q > 0$$

(If 7-day rain is 2 standard deviations above the mean and the runoff formula Q is positive).

***Q*:** Runoff (water that stays on the surface to cause a flood).

P: Rainfall (usually 7-day cumulative for agriculture).

S: Potential Maximum Retention (how much the "soil sponge" can hold). Calculated as:

$$S = \frac{25400}{CN} - 254$$

I_a**:** Initial Abstraction (water soaked up by plants/cracks before runoff starts).
Standard rule: $I_a = 0.2S$

CN**:** Curve Number (e.g., 60 for forest, 85 for **cultivated agricultural croplands**). More info

Feature	Urban Flood (Pluvial)	Agricultural Flood (Rural)
Trigger Metric	**Intensity** (Daily/Hourly)	**Accumulation** (7-10 Days)
Quantitative Threshold	> 95th Percentile of Daily Rain	> 200% of weekly mean + Soil Saturation
Physical Cause	Impervious Cover (Concrete/Asphalt)	Soil Saturation (Sponge is full)
Scientific Basis	**Panthou et al. (2018)**	**Tramblay et al. (2021)**
Standard Agency	**World Bank Urban Handbook**	**African Risk Capacity (ARC)**

- **Douglas, I., et al. (2008). "Unjust waters: climate change, flooding and the urban poor in Africa."**
  - *Key Finding:* This is a foundational paper for African urban flooding. It defines urban floods not just by rain volume, but by the **failure of local infrastructure** to handle "brief but intense" storms. It's perfect for justifying why you are separating urban districts.
- **Adegun, O. B., et al. (2020). "Urban Flooding in Africa: A Review of Challenges and Opportunities."**
  - *Key Finding:* This paper provides a continent-wide review. It specifically mentions that in African cities, a flood is often triggered when rainfall exceeds the **2-year return period rainfall event**, because most African urban drains are only designed for that capacity.
- **Jha, A. K., et al. (World Bank, 2012). "Cities and Flooding: A Guide to Integrated Urban Flood Risk Management."**
  - *Key Finding:* This is a massive technical guide. It defines the **"design standard"** logic. You can cite this to say: *"An urban flood is defined as an event where rainfall exceedance probability is higher than the localized drainage design capacity (usually <10% probability)."*
- **Tshililo, V. P., et al. (2022). "Assessment of Flood Risk and Vulnerability in Rural Communities."**
  - *Key Finding:* Focuses on how rural flooding is often a **"slow-onset"** disaster compared to urban flash floods. It supports your 7-day accumulation logic.
- **Nkiaka, E., et al. (2021). "Mapping flood prone areas and vulnerability in the Lake Chad Basin."**
  - *Key Finding:* This study uses **Multi-Criteria Decision Analysis (MCDA)**. It shows that in rural Africa, the "Flood Trigger" is a combination of **Elevation (Slope)** and **Soil Type**. It supports your "Infiltration/Percolation" requirement.
- **African Risk Capacity (ARC) - "Flood Model Technical Specifications" (2020/2021).**
  - *Key Finding:* This is a **policy document** used by African governments. It defines the "Agricultural Flood" using a **Runoff Index**. It uses a "Relative Threshold" where a flood is defined if the daily runoff is in the **top 10% of historical events** for that specific farm-pixel.
- **World Bank Climate Change Knowledge Portal (CCKP):**
  - *Reference:* They provide "Climate Risk Country Profiles" for every African nation.
  - *Technical Metric:* They define "Heavy Precipitation Days" as days where rainfall is **>95th percentile** of the 1981–2010 baseline. This is your most "citeable" standard for your Admin 2 districts.
- **WMO (World Meteorological Organization) No. 1103: "Guidelines on Analysis of extremes."**
  - *Key Finding:* This provides the international standard for the **"R95p" index** (annual total precipitation from days >95th percentile). You can tell your boss: *"We are following the WMO R95p standard to define our flood days."*
- **Sylla, M. B., et al. (2015). "Climate change over Africa: Challenges and opportunities."**
  - *Key Finding:* This paper discusses the **"Climatological Mean"** in Africa. It supports the logic that in dry regions, a flood might be 50mm, while in wet regions, it might be 200mm—therefore, the "Percentile" is the only fair metric.

"To identify and extract specific Admin 2 districts where "Agricultural Flooding.

Data Source

- **Global Administrative Areas (GADM 4.1) Administrative Level districts.**
- **Global Land Analysis and Discovery (GLAD) - 30m - Landsat time series (Potapov et al., 2021).**
- **CHIRPS Daily (Climate Hazards Group InfraRed Precipitation with Station data)**
- **ECMWF ERA5-Land Daily Aggregated**

Seasonal Reanalysis

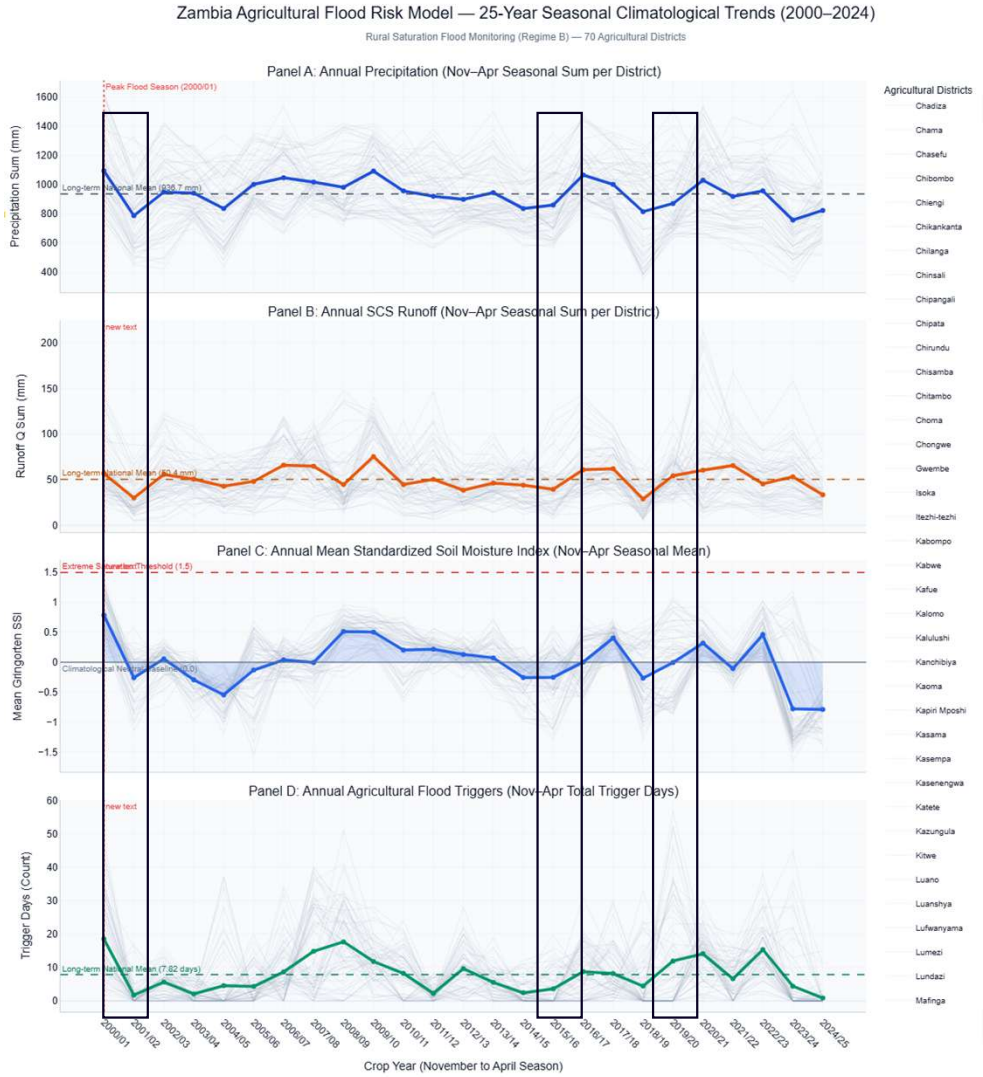
Panel A (Precipitation): Tracks the seasonal **Meteorological Supply** (Nov–Apr). It establishes the national baseline (~936.7 mm) and identifies the raw inter-annual variability of water inflow.

Panel B (Runoff): Tracks the **Physical Rejection** of water using the SCS Curve Number (CN=85). This panel is now hydrologically consistent (discrete daily sum), showing that ~5% of seasonal rain translates into direct surface runoff.

Panel C (SSI): racks **Soil Memory** using the Gringorten non-parametric index. Positive values (Blue) indicate a "Full Sponge," which pre-conditions the landscape for saturation flooding.

Panel D (Triggers): The final insurance product. It counts days per year meeting the **Dual-Trigger Criteria** ($SSI \geq 1.5$ AND $Runoff \geq 20mm$). This isolates catastrophic events from standard rainy days.

The **2000/01 season** represents the maximum systemic flood risk (avg. 18 trigger days), while the **2023/24 El Niño** marks the record low, showing zero flood triggers.



Validation:

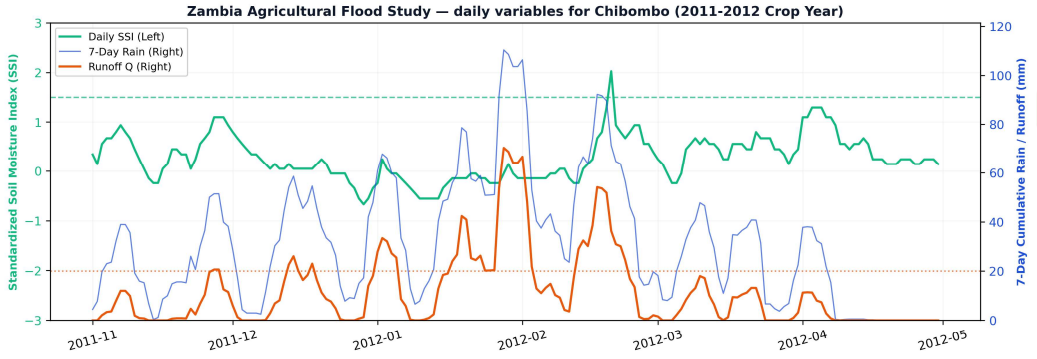
Goal: To perform a daily-level validation of the dual-trigger mechanism during three historically catastrophic flood seasons: **Chibombo (2011/12)**, **Kapiri Mposhi (2007/08)**, and **Monze (2007/08)**.

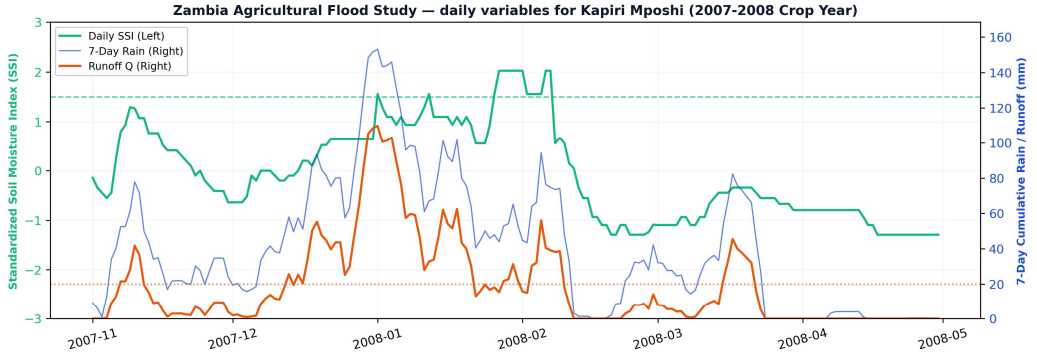
Data:

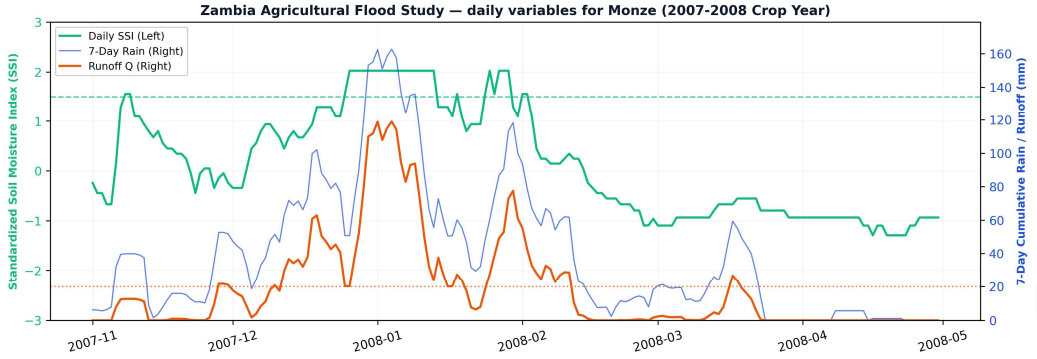
Integration: Overlays **Rainfall** (Blue), **Runoff** (Orange), and **Soil Saturation** (Green).

The "Collision" Logic: Red 'X' markers identify where **Runoff ≥ 20mm** and **SSI ≥ 1.5** collide.

Hydrological Selectivity: Captures late-season "Saturation Floods" while filtering out early-season heavy rain on dry ground.







Data Pre-processing (The Baseline)

Step 1: Data Pre-processing (The Baseline)

1. Collect 30 years of daily rainfall (CHIRPS data).
2. Calculate the **95th Percentile (P_{95})** for each pixel.
3. Calculate the **Weekly Mean (μ)** and **Standard Deviation (σ)** for each month.

Step 2: Spatial Masking (The "Where")

1. Use **ESA WorldCover 10m**.
2. Calculate the % of **Class 50 (Built-up)** in Admin 2 district.
3. If **Built-up > 10%**, use the **Urban Formula**.
4. If **Built-up < 10%**, use the **Agricultural Formula**.

Step 3: Calculating the Event (The "When")

- **For Urban Districts:** Look for days where daily rain > P_{95}
- **For Agricultural Districts:**

Look at the **Soil Moisture Index (SMI)** from the last 7 days.

If SMI is high (>0.7), calculate Q using the **SCS Formula**.

If Q is significant (e.g., >20mm of surface water), mark as a flood.

To identify and extract specific Admin 2 districts where "Urban Flooding"

Focus on Imperviousness and Runoff

Problem:

Occur in small, dense urban "hotspots." Even a district that is only 10% built-up can experience 100% urban flood destruction.

Study Area Data Source & Target Class

- **Utilizing ESA WorldCover (Global Land Cover at 10m resolution).**
- **Target Class:** Class 50 (Built-up). Land covered by buildings, roads, and other man-made structures.
- High-resolution **10m data** allows us to calculate the exact percentage of "sealed surfaces" within an **Admin 2 boundary**, which is the primary driver of flash floods.
- **GHSL (Global Human Settlement Layer).**

Extracting Districts via the Impervious Cover Model (ICM)

- **The Rule:** Extract any Admin 2 district where Class 50 (Built-up) exceeds **10% of the total land area (***Schueler, T. R., et al. (2009***).**
  - At **10% impervious cover**, the natural water cycle is permanently altered.
  - Surface runoff increases by 2x to 5x compared to forested land.
  - Stream health begins to decline, and flood frequency increases sharply.

The UN DEGURBA Standard

- **The Rule:** A district is extracted if it meets the **Degree of Urbanization (DEGURBA)** criteria (*European Commission (2021). Applying the Degree of Urbanization – A methodological manual*):
  - Urban Center: High-density clusters with a population of at least 50,000.
  - Built-up density: Pixels in the ESA map show a density of >50% within a 1km² area.
- It is the official method endorsed by the UN Statistical Commission to define "Cities" vs. "Rural areas."

Extraction Strategy for Admin 2 Flood Study

- Adopt Strategy 1 (The 10% Impervious Threshold) as the primary extraction rule.
- Integrate Strategy 2 (DEGURBA) to categorize the resulting districts into "Cities" or "Towns."
- **The "Tipping Point" Logic (**Schueler (2009) – The Impervious Cover Model (ICM)):
  - In urban hydrology, 10% impervious cover is the globally recognized "tipping point."
  - At this threshold, natural drainage is compromised, and flood frequency/volume increases significantly.
  - The World Bank Urban Flood Risk Handbook (2023) recommends using land-cover density to prioritize flood interventions in the Global South/Africa.

Proposed Methodological

- **Overlay:** Intersect the **ESA WorldCover (Class 50)** raster with the **Admin 2 Vector layer.**
- **Calculate:** Run zonal statistics to find the percentage of "Built-up" area per district.
- **Apply Filter:** Extract all districts meeting **either** condition:
  - **A:** Total Built-up Area > 10% (Hydrological impact).
  - **B:** Contains a cluster with >50% density (Urban center impact).
- **Validate:** Cross-reference with historical flood records in Africa.

Tramblay et al. (2021). "Hydrological floods in Africa." (Justifies why land-cover change is the primary driver of flood risk)

- **Scientifically Rigorous:** Uses the 10% "Tipping Point" threshold (Schueler, 2009).
- **Policy Aligned:** Follows the UN-standard DEGURBA framework for defining urbanity.
- **Context Specific:** Acknowledges the unique sprawl patterns in African cities (Abass et al., 2018).
- **Actionable:** Provides a clear, mathematical cutoff for your study area (10% Built-up = Urban Study Area).

References:

- Abass, K., Appiah, D.O. and Afriyie, K. (2018). Urban sprawl and flooding in Kumasi, Ghana: driving forces, social-relative causes and households' adaptation strategies. *Environmental Development*, 26, pp.54–68. doi:10.1016/j.envdev.2018.03.006.
- Angel, S., Blei, A.M., Parent, J., Lamson-Hall, G. and Galarza Sanchez, N. (2016). *Atlas of Urban Expansion – 2016 Edition*. New York: New York University, Nairobi: UN-Habitat, and Cambridge, MA: Lincoln Institute of Land Policy.
- European Commission, Statistical Office of the European Union, FAO, UN-Habitat, OECD and World Bank. (2021). *Applying the Degree of Urbanisation: A methodological manual to define cities, towns and rural areas for international comparison*. Luxembourg: Publications Office of the European Union.
- Schueler, T.R., Fraley-McNeal, L. and Cappiella, K. (2009). Is Impervious Cover Still Important? Review of Recent Research. *Journal of Hydrologic Engineering*, 14(4), pp.309–315. doi:10.1061/(ASCE)1084-0699(2009)14:4(309).
- Tramblay, Y., Villarini, G. and Mumba, M. (2021). Stronger evidence for the role of land-use change and human activities on increased flood risk in Africa. *Environmental Research Letters*, 16(2), p.024028. doi:10.1088/1748-9326/abd16e.
- Zanaga, D., Van De Kerchove, R., De Keersmaecker, W., Souverijns, N., Brockmann, C., Quast, R., Wevers, J., Grosu, A., Paccini, A., Vergnaud, S., Cartus, O., Santoro, M., Fritz, S., Lesiv, M., Arino, O. (2021). *ESA WorldCover 10 m 2020 v100*. doi:10.5281/zenodo.5571936.



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